Paleomagnetic evidence for the extent of Mackenzie igneous events

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North-northwesterly striking Mackenzie diabase dikes of middle-Proterozoic (Helikian) age are profuse in the western part of the Canadian Shield. Published paleomagnetic data on dikes of this trend in Mackenzie District, on the Muskox Intrusion, the Coppermine River volcanic rocks, and the Sudbury dikes suggest that they are all products of closely related igneous events. This paper presents paleomagnetic data that suggest that the intrusion of extensive diabase sheets in the East Arm of Great Slave Lake, and of dikes as far to the northeast as Melville Peninsula and as far to the southeast as Manitoba, were also parts of these events. The mean paleomagnetic pole position for the Mackenzie dikes and for related intrusive and extrusive rocks is 3½° N, 171° W with $A_{95} = 4½°$. Radioactive age determinations, some of which are unpublished, indicate an age of about 1200 m.y. for the formation of these rocks. It is suggested that for convenience all of these apparently related intrusive and extrusive igneous episodes be referred to as Mackenzie igneous events.

Introduction

The many swarms of diabase dikes intruding the Canadian Shield have been distinguished largely on the bases of trend and potassium–argon dating (Fahrig and Wanless 1963). The Mackenzie swarm is the most spectacular, sweeping from the Arctic coast of the western Canadian Shield southeasterly towards northwestern Ontario. Preliminary paleomagnetic work on thirty-four samples from twenty-three dikes suspected of being members of the Mackenzie swarm was reported by Fahrig *et al.* (1965). The directions of magnetization of the samples were generally similar, supporting the hypothesis that these dikes were intruded at approximately the same time. Robertson (1964, 1969) studied the paleomagnetism of rocks of the Muskox Intrusion, of Mackenzie dikes adjacent to the intrusion, and of the overlying Coppermine River volcanic rocks. He pointed out that there is geological evidence suggesting that these rocks are "part of a single Proterozoic phase of igneous activity", and his paleomagnetic results provide further evidence. The mean paleomagnetic pole position calculated by Robertson (1969) (4.7° N, 168.8° W) lies close to the preliminary pole position calculated by Fahrig *et al.* (1965) for the Mackenzie dikes (3.8° N, 177.3° W). The intent of the present work was to corroborate preliminary paleomagnetic data from the Mackenzie dikes, to determine the areal extent of this swarm, and to test the hypothesis that the diabase sills in the East Arm of Great Slave Lake also form part of Robertson's (1969) "single Proterozoic phase of igneous activity".

Sampling

Seventy-one cores were collected at fifteen sites from four areas of the Canadian Shield (areas A, B, C, D of Fig. 1) by means of a diamond drill and were oriented by means of a solar compass.

Mackenzie swarm representatives were sampled at two sites in area A to corroborate previous paleomagnetic data on dikes of this swarm. Samples from sites 3 to 6 (area B on Fig. 1) from northeastern Manitoba and sites 7 to 9 (area C on Fig. 1) from the Melville Peninsula (Heywood 1967) were collected to examine the possible extension of the Mackenzie dikes to the southeast and east–northeast respectively. Trend, general similarity of potassium–argon ages, and the geochemical nature of the dikes in areas B and C suggest that they are related to the Mackenzie dikes.

Sites 10 to 15 from area D, Fig. 1, are from sheets of diabase that outcrop in the East Arm

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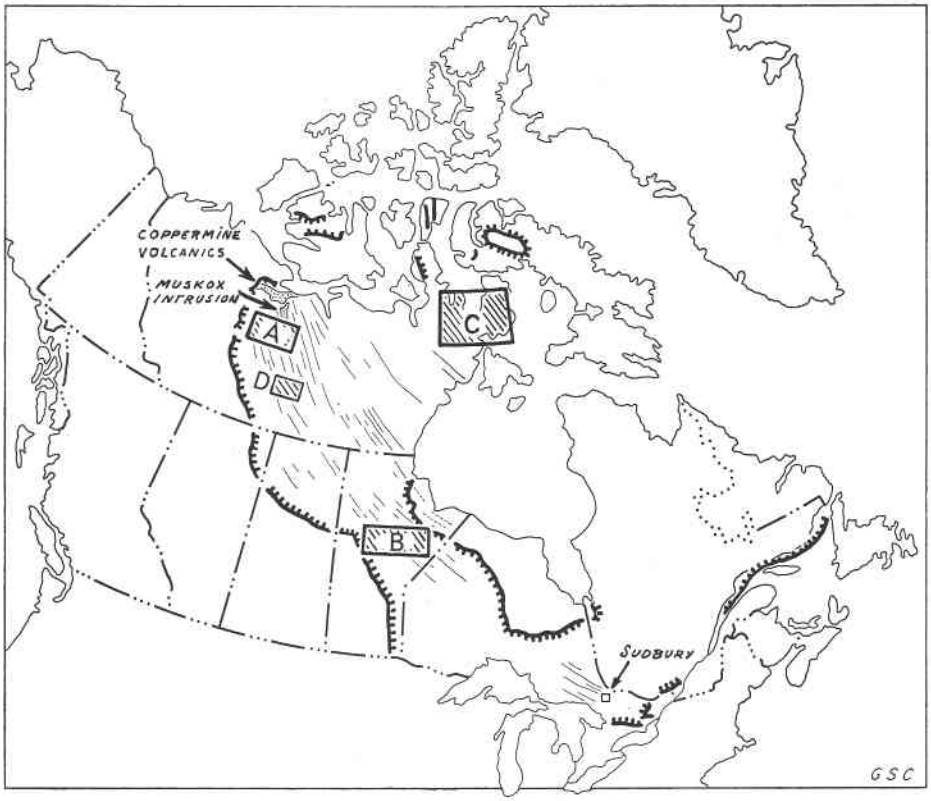


FIG. 1. Index map showing sampling areas and schematic representation of Mackenzie and Sudbury dikes.

of Great Slave Lake. These rocks were described by Stockwell (1936) as follows: "Diabase dikes and sills cut the Et-then series and all older rocks and are later than the large faults. The sills are up to 500 feet or more thick, up to 95 miles long and occur in the gently dipping rocks of the north part of the area. In the more complexly folded rocks of the south part of the area, the diabase characteristically takes the form of moderately dipping dykes of irregular trend." Sites 10, 11, 14, and 15 are from the gently dipping sills described by Stockwell (1936), and sites 12 and 13 from his moderately dipping dikes of irregular trend. Suspicion of a close relationship between these sills and the Mackenzie dikes was raised by their proximity and by potassium–argon age determinations. Few dikes cut the sheets and it was felt that the dikes may have acted as feeders to the sheets.

The cores are 2.5 cm in diameter, and two specimens 2.5 cm in length were cut from each core. Each core represents an independent sample and in the statistical calculations the results from the two specimens were combined and the mean result taken to represent the sample.

Measurements and Results

The laboratory work was carried out in the paleomagnetic laboratory of the Geophysics Division of the University of Alberta. The natural remanent magnetization (NRM) of each specimen was determined with a Princeton Applied Research Corporation spinner magnetometer and site mean NRM (natural remanent magnetization) directions of magnetization and related data of each site are given in Table I and Fig. 2*a*. Good with-in site grouping was observed at twelve of the fifteen sites and the site mean directions (except site 11) form a group away from the direction of the present earth's field.

The specimens were cleaned magnetically

TABLE I

Site directions for Mackenzie dikes A, dikes in Melville Peninsula C, and north-eastern Manitoba B, and the East Arm diabases D

Area	Site No.	Site position		M_n						M_{ac}					
		Lat. N	Long. W	N	D°	I°	R	k	B_{ac}	N	D°	I°	R	k	α_{95}
A	1	65.10	117.60	6			3.210*		20	4	233	25	3.982	169	5.4
	2	65.80	111.10	4	240½	38	3.968	94	20	4	235½	22½	3.989	280	4.2
	3	55.80	95.85	6	264	65½	5.937	79	15	6	252½	52½	5.898	49	8.2
	4	54.80	94.25	6	264½	37½	4.075	3	10	5	249	28	4.878	33	11
B	5	54.70	95.85	3	268	47	2.950	40	15	3	261	25	2.925	27	16
	6	55.45	98.40	5	257	44½	3.508	3	10	4	264½	28	3.982	165	5.5
	7	66.35	87.65	5	271	43	4.918	49	20	5	261	17½	4.998	1530	1.6
C	8	67.05	87.25	5	265	68	4.803	20	30	5	267½	26	4.762	17	15
	9	68.45	82.20	5	271½	49	4.765	17	10	5	267½	10½	4.969	128	5.5
	10	62.70	110.35	6	246	33½	5.971	170	15	6	243	18	5.991	570	2.4
	11	62.70	109.85	4	272½	82½	3.964	83	15	4	201	85	3.960	74	8.1
	12	62.55	109.15	4	236½	58	3.923	39	15	4	237½	32½	3.986	212	4.8
D	13	62.45	110.65	4	248½	55	3.907	32	20	4	239	41	3.982	164	5.5
	14	62.55	111.50	4	240½	36½	3.964	83	20	4	238½	19½	3.997	1030	2.2
	15	62.80	110.75	4	247½	33	3.997	900	15	4	245½	21	3.995	626	2.8

N = number of samples measured.
R = length of the resultant of the N unit vectors.
D = declination of remanent magnetization.
I = inclination of remanent magnetization (all positive).
M_n = NRM.
M_{ac} = RM after demagnetization in alternating field B_{ac} (mWb m^{-2}).
$k = (N - 1)/(N - R)$ = estimate of Fisher's (1953) κ parameter.
α_{95} = semiangle of cone of 95% confidence for mean direction.
*This number refers to the 'resultant' of data, which are random on Watson's (1956) criterion.

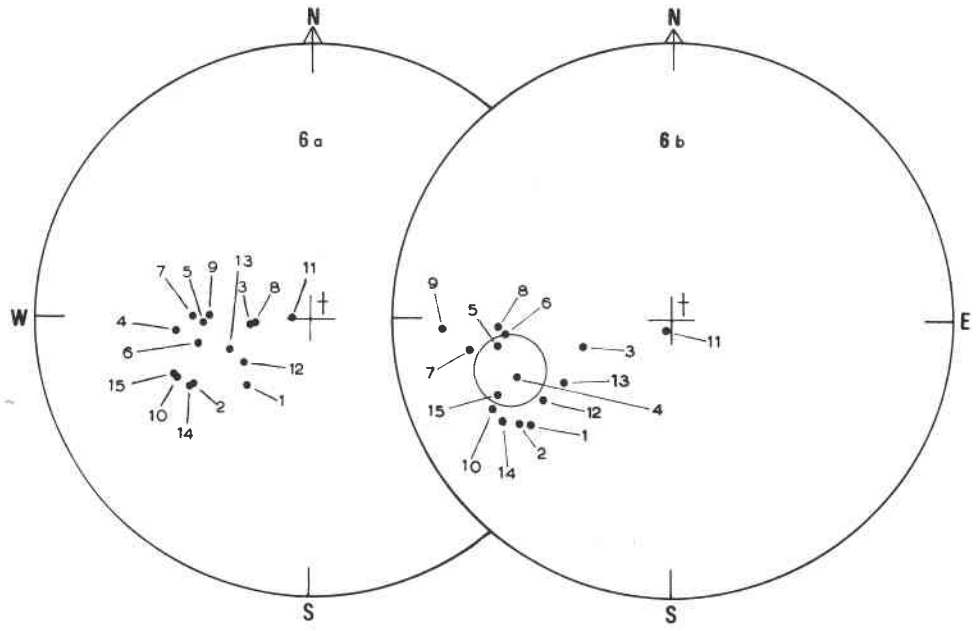


FIG. 2. Stereographic projection of the site mean directions of magnetization. (a) NRM directions; (b) directions of remanent magnetization after magnetic cleaning. Solid line circle shows 95% confidence for the (recomputed) mean direction obtained by Fahrig *et al.* (1965). Solid circles = north-seeking directions down. † = present earth's field, north-seeking down.

using the procedure outlined by McElhinny and Gough (1963). Thus at least two specimens from each site were subjected to successively higher alternating magnetic fields in order to establish an optimum demagnetizing field. The (peak) field for the test specimens was increased in 5 mWb/m² (50 gauss) steps to 30 mWb/m², then in 10 mWb/m² steps to fields in excess of 50 mWb/m². The majority of the test specimens exhibited a stable end-point beginning in fields of about 10–15 mWb/m²; typically they showed random changes in magnetization directions after subjection to fields of 40–50 mWb/m². The remainder of the specimens from each site were then treated in the optimum field thus obtained for that site. The relevant data are given in Table I. Data from four of the cores (involving three sites) were excluded from the calculation of the cleaned site means because they were thought to have been remagnetized by lightning strikes. The NRM intensities of magnetization of the specimens from an excluded core were ten to one hundred times greater than the intensities of the remaining specimens from the site. The NRM directions of the specimens from an excluded core were generally well away both from that of other specimens at the same site and also from that of the present earth's field. A test specimen from each of the excluded cores, demagnetized in steps, was found to retain its initial direction of magnetization in fields considerably in excess of the optimum field for the site. The subsequent changes were generally large and random.

The scatter of the magnetization directions of the samples from each site was either decreased, or not significantly altered by the demagnetization procedure and the mean magnetization direction of each site, with the exception of site 11, moved away from the direction of the present earth's field. The end-points indicated by the test specimens, the reduction of scatter of the with-in site directions after cleaning and the movement of the site mean directions away from the direction of the present earth's field, all indicate that a viscous component was removed at lower fields leaving a stable component, which is interpreted as the primary magnetization acquired by the samples during the initial cooling of the basaltic magma.

There is abundant evidence to support this contention. The margins of all the intrusive bodies consist of narrow chill zones made up of unaltered micro-phenocrysts of plagioclase and clinopyroxene in an opaque matrix. The opaque matrix would have recrystallized to form identifiable minerals characteristic of the greenschist facies under conditions of even low-grade regional metamorphism. It is also extremely unlikely that the entire western Canadian Shield has undergone a thermal event so synchronized that all of these basic rocks were remagnetized in a new and consistent pattern. Furthermore we have measured magnetization directions of diabase specimens from the District of Mackenzie that yield K–Ar ages several hundred million years greater than the age attributed to the Mackenzie igneous events (see below) and have found that the older diabase has magnetization directions quite different from those of Mackenzie rocks. These older rocks must have undergone any remagnetization phenomena experienced by neighboring Mackenzie rocks, yet they retain directions of magnetization quite different from those of the Mackenzie. Finally, Robertson (1969) has provided contact evidence from the Muskox Intrusion that indicates that this body has a stable component of magnetization formed during initial cooling of the intrusive mass.

The possibility of post-magnetization tilting of the dikes and sheets cannot be directly evaluated, as these bodies lack a reference plane to indicate their primary orientation. It is observed, however, that with one exception site poles over this huge area are closely grouped suggesting that the assumption that no local tilting took place is valid. Furthermore, the size of the area is such that it is difficult to visualize movement of the entire cratonic plate in such a way as to significantly tilt the collection sites as a group.

Figure 2*b* shows the site mean directions after demagnetization; the grouping noted in NRM has been largely maintained. The mean direction of magnetization at site 11 is clearly anomalous when compared with the other five sites from the East Arm diabase, although the with-in site precision parameter k is similar to the other sites. Site 11 comprises a large outcrop of diabase on the lake shore, which may have rotated as a single unit, either tectonically, or more likely as a result of a gravity initiated

TABLE II

Results of detailed alternating field demagnetization of site 2 (symbols are the same as for Table I)

B_{ac}	D°	I°	k
0	240	38	94
15	236	24	226
20	235	23	280
30	237	20	136
40	237	21	78

slide. Site 11 has been excluded from between-site and inter-area considerations.

In earlier work on the Mackenzie dikes Fahrig *et al.* (1965) subjected their specimens to alternating field demagnetization in 8 mWb/m² (peak). Our site 2 was analyzed in considerable detail to determine whether this treatment was adequate. The results from this site (Table II and Fig. 3) suggest that although Fahrig *et al.* (1965) did not employ the optimum field for a.f. washing, their Mackenzie dike mean directions are unlikely to be significantly in error.

As it is more meaningful to compare site geomagnetic pole positions rather than site mean directions of magnetization when the sites

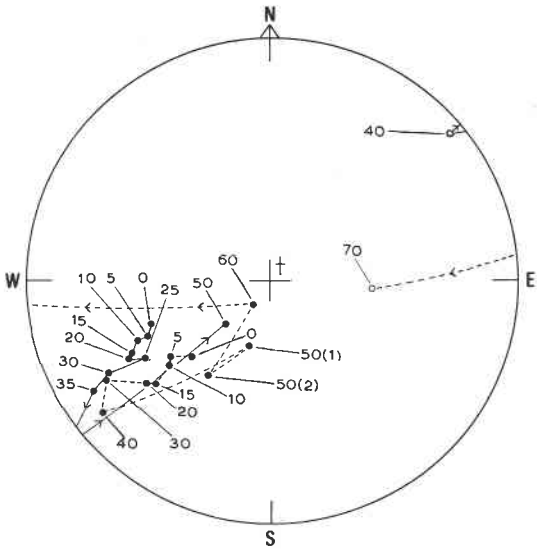


FIG. 3. Lower hemisphere stereographic projection showing the response of two test specimens from site 2 to step-wise alternating field demagnetization. Fields given in mWb/m². Solid circles = north-seeking directions down. Open circles = south-seeking directions down. † = present earth's field, north-seeking down.

are distributed over a large area, pole positions were calculated for each site (Table III and Fig. 4). They are very closely grouped. Mean pole positions for each of the four areas of Fig. 1 are also given in Table III. They form a close group and their error circles (95% confidence) overlap except for those of areas A and C of Fig. 1. The original data of Fahrig *et al.* (1965) have been reanalyzed and the mean pole position for the Mackenzie dikes recalculated giving unit weight to sites (= dikes) rather than to samples. In addition, the results from one dike were omitted as this dike is thought to be a member of an older swarm of dikes. The error circle for the recalculated mean pole intersects the circles calculated for dikes in areas B, C, and D and touches that for A. The paleomagnetic results, therefore, strongly support the hypothesis that the intrusions sampled in areas B, C, and D are closely related to the Mackenzie dikes. A mean pole position for the fourteen acceptable sites of the present work is recorded in Table IV.

Mackenzie Igneous Events

Paleomagnetic evidence suggests that the Mackenzie dike swarm is even more extensive than was previously thought and includes dikes of similar trend in the Melville Peninsula and in northeastern Manitoba; it also suggests that the East Arm diabase is related to this dike swarm. Table IV summarizes the paleomagnetic data from (*a*) the present work, (*b*) the Mackenzie dikes (Fahrig *et al.* 1965), (*c*) the Muskox Intrusion, Coppermine River volcanics and Mackenzie dikes (Robertson 1969), and (*d*) the Sudbury dikes (Larochelle 1967). The data of (*c*) Table IV have been taken from Robertson's manuscript and are combined results in which each of the three major petrologic units of the Muskox Intrusion, the Coppermine River volcanics, and the dikes have been given unit weight. Row (*d*) Table IV, is the result of a detailed study of the Sudbury swarm. Early preliminary paleomagnetic work on this swarm was published by Strangway (1961, 1964), Sopher (1963), and Fahrig *et al.* (1965). In the last contribution it was suggested on paleomagnetic evidence that the Sudbury swarm could be correlated with the Mackenzie swarm, a conclusion compatible with other evidence (trend, radioactive age

TABLE III

Site pole positions

Area	Site	Site pole Lat.	Site pole Long.	Area pole position Lat.	Area pole position Long.	K	A_{95}
A	1	2° S	169° W	2° S	167° W		
	2	3° S	165° W				
B	3	18° N	153° W	9° N	165° W	41	10°
	4	1° N	159° W				
	5	5° N	171° W				
	6	9° N	175 °W				
C	7	5° N	166° W	7° N	168° W	310	4°
	8	11° N	170° W				
	9	4° N	168° W				
D	10	3° S	172° W	0° N	168° W	120	5°
	12	2° N	163° W				
	13	8° N	163° W				
	14	5° S	169° W				
	15	1° S	174° W				

K = precision of site poles
A_{95} = semiangle of the cone of 95% confidence for the mean pole position.



FIG. 4. Site pole positions.

determinations and, to a certain extent, geochemistry).

The agreement between the four sets of data in Table IV is remarkable (see Fig. 5), and is consistent with the idea that the igneous events referred to above are closely related in time. There is now a considerable body of paleomagnetic data supporting the idea of an immense single Proterozoic period of basic igneous activity which encompasses the Mackenzie dikes, the Sudbury dikes, the Muskox Intrusion, the Coppermine River volcanics, and the East Arm diabase sheets. The mean pole position obtained by combining the data of Table IV is 3.5° N, 170.8° W with K = 192 and A_{95} = 4.5°. This is an extremely reliable pole position relative to the western Canadian Shield for the time of the Mackenzie igneous events.

An interesting aspect of the paleomagnetic results is the apparent absence of reversals of polarity shown by any of the Mackenzie rocks. Robertson (1964, 1969) found no reversals in 24 sites, which represent more than 5000 ft of Coppermine River volcanic flows, in several thousand feet of the layered Muskox Intrusion, or in dikes near the intrusion. No reversals

TABLE IV

Pole positions from some related Precambrian rocks of Canada (symbols as in Table III)

Study	N	Pole Position Lat.	Pole Position Long.	K	A_{95}
(a) Present work	14 sites	3.5° N	166.9° W	82	4
(b) Mackenzie dikes	22 dikes	7.8° N	176.6° W	8	10
(c) Muskox etc.	5 units	4.7° N	168.8° W	93	8
(d) Sudbury dikes	11 sites	2.0° S	171.0° W	210	3



FIG. 5. Circles of 95% confidence for the mean pole positions for (a) the present work, which includes the data from sites in areas A, B, C, and D of figure 1, (b) the Mackenzie dikes (Fahrig *et al.* 1965), (c) the Muskox Intrusion etc. and (d) the Sudbury dikes.

were observed in samples of 22 Mackenzie dikes previously studied (Fahrig *et al.* 1965) or in samples of nine dikes and several sills reported in the present study. There are two situations that independently or together can explain this phenomenon. The intrusive and extrusive activities may represent a brief period of time, reversals may have occurred infrequently around the time of these events, or both situations may have existed. There is no direct evidence of the maximum duration of Mackenzie events; however, other flood basalt provinces have developed over millions, or tens of millions of years (Clifford 1968) so the great extent of the Mackenzie rocks and the thickness (up to 10 000 ft) of the Coppermine River volcanics suggest that at least several million years were involved in their formation. During this period of at least several million years the earth's field apparently maintained a constant polarity.

The Age of the Mackenzie Igneous Events

It is important to establish the age of these events as accurately as possible and towards this end a considerable body of potassium–argon radio-isotopic data have been accumulated.

Mackenzie Dikes

Ages obtained by the Geological Survey of Canada on Mackenzie dike material during the period 1963–67 are shown in Table V. The average age obtained on chilled dike material or biotite concentrates in 1963 is 1317 m.y., and for subsequent determinations (1964–1967), 1023 m.y. The greatest age obtained in the 1963 analyses was 1360 m.y.; for later determinations, 1200 m.y. Ordinarily, the greater ages would be expected to provide the best estimate of age, as argon loss during the past billion years is likely to have decreased the potassium–argon ages, particularly in the analysis of whole-rock samples. However, analytical techniques in use at the Geological Survey have been improved since the 1963 work, so the later (1964–67) analyses are considered more reliable. The greatest ages indicated by these later analyses are in the 1100–1200 m.y. range, and these provide the best estimate of the age of these dikes as given by this set of data. Younger basic igneous rocks may contain enough primary argon, that is, argon brought up from the mantle, to significantly increase potassium–argon ages. The rocks discussed here, however, are so old that any primary argon is probably entirely masked by the argon generated since their emplacement.

Leech (1966) has reported eleven potassium–argon determinations on N.N.W.-striking dikes of the District of Mackenzie. These dikes probably belong to the Mackenzie swarm and the eleven dates are closely grouped around an average of 1155 m.y. These determinations were made chiefly on chilled contact material, but they include two whole-rock determinations made on coarser dike material and one made on biotite from country rocks from within 2 cm of a dike margin. The oldest of the eleven ages was 1250 m.y. Burwash *et al.* (1963) had previously reported potassium–argon ages in the range 1000–1100 m.y. on samples from N.N.W.-striking dikes in the same region.

Coppermine River Volcanics

Table V gives the Geological Survey whole-rock potassium–argon age determinations on the Coppermine River volcanics. The greatest ages are in the 1100–1200 m.y. range; however, there is considerable scatter in the direction of younger values for these rocks. As there is no evidence of a wide range in the age of volcanism, the lower potassium–argon ages are

TABLE V

Geological survey of Canada K–Ar age determinations

G.S.C. Number[1]	Sample No.	Material analyzed (numbers indicate % minerals in mixed samples)	K–Ar age (m.y.)[4]
		Mackenzie Dikes	
63–43	FA-170-62(3)	biotite (30) hornblende (40) chlorite (20) pyroxene (10)	1245
63–44	FA-114-62(1)	biotite (90)	1360 ± 50
63–48	FA-138-62(3)	whole rock (chilled)	1335
63–59	FA-185-62(2)	biotite (60) chlorite (35)	1315
63–60	FA-124-62(3)	biotite (80)	1330 ± 50
63–88	SDC-60-227A	whole rock (medium grained)	1110
63–89	SDC-231-5	whole rock (medium-coarse)	975
64–39	T-189-62	whole rock (medium grained)	1050 ± 95
64–73	FA-32-63-(2)	biotite concentrate	1110 ± 100
64–78[3]	FA-40-63-(1)	whole rock (chilled)	1105 ± 140
65–57[2]	FA-650041	whole rock (chilled)	1010 ± 135
65-58[2]	FA-650021	whole rock (chilled)	941 ± 100
65–75	DF-A255-64	whole rock (chilled)	899 ± 90
65–102	BA-2-681C	whole rock (medium grained)	1130 ± 135
67–unpub.[3]	FA-24-66	biotite concentrate mixed with chilled whole rock	1095 ± 115
67–unpub.[3]	FA-23-66	whole rock (chilled)	865 ± 99
67–unpub.	DF-A820-66	whole rock (chilled)	938 ± 56
67–unpub.[2]	FA-12-66	whole rock (chilled)	964 ± 112
67–unpub.[2]	FA-14-66	Biotite (70) chlorite (25) hornblende (5)	1005 ± 40
		Coppermine Volcanics	
63–77	FD-105-59	whole rock	735
63–78	FD-286-62	whole rock	918
63–91	SDC-62-3062	whole rock	1200
64–54	FD-106-59	whole rock	863 ± 115
64–55	HF-149-59	whole rock	788 ± 110
64–57	FO-252-59	whole rock	1065 ± 100
64–68	RN-132	whole rock	1100 ± 130
67–unpub.	BLB-6-48	whole rock	767
67–unpub.	BLS-1-66	whole rock	890
67–unpub.	BLS-161-66	whole rock	769
		Muskox Complex	
60–38	SDC-59-1177	biotite (associated with sulphide intergrowths)	1155
63–90	SDC-62-3087-G	whole rock (chilled feeder dike)	1095
		East Arm Sheets	
67–unpub.	FA-21-66	biotite (65) hornblende (35)	1020 ± 85
		Sudbury dikes	
65–117	FA-202-62	biotite (95) Impurities (chlorite, hornblende 5)	1330 ± 50
64–107	FA-85-63(3)	Biotite concentrate (with pyroxene, feldspar, amphibole)	950 ± 75
64–99	FA-55-63(1)	Biotite (90+)	1240 ± 60

NOTES: 1. Analytical details for these samples are published in a series of annual papers published by the Geological Survey of Canada. Those analyses with numbers beginning with 60–, are published in Geological Survey of Canada paper 61–17. Those numbered 61–, are published in paper 62–17, and so on.
2. These analyses were carried out on material from dikes in area C (Fig. 1).
3. These analyses were carried out on material from dikes in area B (Fig. 1).
4. The errors quoted are the limits of laboratory error (95% confidence).

thought to reflect argon loss. Fewer low numbers have been obtained from whole-rock chilled diabase than from samples of the Coppermine River lavas, indicating that argon loss was greater from extruded basalt than from the equivalent intrusive material. However, the Mackenzie dikes that have been dated are some distance south and east of the Coppermine River volcanics, so it is probable that the volcanics have been affected by a young thermal event that has not significantly affected the dikes. Preliminary results from a Geological Survey rubidium–strontium study of the Coppermine River volcanics gives an age of about 1200 m.y. (R. K. Wanless, personal communication). This confirms the age indicated by the higher potassium–argon numbers.

Muskox Intrusion, East Arm Sheets, and Sudbury Dikes

The Geological Survey has obtained two potassium–argon dates on the Muskox Intru-

sion, one on the East Arm sheets and three from what are undoubtedly Sudbury dikes (Table V). These are within the range of dates obtained by the Geological Survey, Leech (1966), and Burwash *et al.* (1963) for the Mackenzie dike swarm. Fairbairn *et al.* (1960) reported a potassium–argon age of 1020 m.y. for biotite of a granophyric segregation in a Sudbury dike, while van Schmus (1964) has obtained a rudibium–strontium age of 1205 ± 50 m.y. for the Sudbury dikes.

The dating programs reviewed above suggest the single period of basic magmatic activity which gave rise to the Mackenzie and Sudbury dikes, the Muskox Intrusion, the Coppermine River volcanics, and the East Arm sheets occurred about 1200 m.y. ago. This indicates that movement on several major faults in the region of the East Arm diabase (Stockwell 1936) is older than 1200 m.y.

Discussion

The spacial relationships of the basic intrusive and extrusive rocks discussed above, parallelism of the trends of those intrusions that have dike form, reasonable consistency in the K–Ar dating and very close agreement of paleomagnetic data all suggest that the Mackenzie igneous events were related in time. The rocks resulting from these events are spread throughout a million square miles of the western Canadian Shield, and when one considers the possibility that the Coppermine River volcanics may be a minor erosional remnant of much more widespread surface and near-surface basaltic material these events must represent the passage of enormous quantities of basalt from mantle to crust.

The very dimensions of these events tends to define the mechanism that caused them. The presence of a great swarm of dikes sweeping across the shield from the Arctic coast to the Sudbury region indicates the presence of a zone of tension or of low pressure and it indicates the availability of molten basalt below the crust of this zone. This situation seems by analogy (Clifford 1968) to be most similar to the situation existing along the oceanic ridges. The formation of the Mackenzie dike swarm may well have been a response to the same type of activity that causes ocean floor spreading. In the case of continental dike swarms the rifting process apparently ceased at an early stage, and aside from minor spreading indicated by the total thickness of the dikes across the swarm, continental break-up did not occur.

If ocean floor spreading is a response to mantle convection, then continental dike swarms may be due to either weak or to very short lived convection currents. An effect similar to the second of these possibilities would obtain if a rising current shifted rapidly laterally relative to the overlying continental crust. The possibility of such a shift might be investigated by examining various parameters of the dikes across a wide part of the swarm.

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